

AZAD INDIA MOBILITY LIMITED
Preventive Maintenance Service Checklist
Check Sheet (C) - 80 000 KMs

Date of Inspection		Vehicle Reg. No	
Depot		Odometer Reading	

Status mark	(v) OK	(RR)-Repair Recommended	(RI)-Repair Immediately	
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Max. Labour Time	Start Time	End Time	Total Time
Minutes			

Sr. No	Check Parameter	Inspection Method	STD Time (Minutes)	Status	Remarks
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I. Check Bus Exterior

1	All Lamp Glasses both Front/Side / Rear	Visual			
2	All Flap Door Lock and Shockers, LHS	Visual			
3	Rear Flap Door Lock and Shockers	Visual			
4	All Flap Door Lock and Shockers, RHS	Visual			
5	Function of Hydraulic & Screw Jacks	Functional			
6	Driver Door Window Glass/ Door Locks	Visual			
7	Uneven body level, one side or both the side	Visual			
8	Check the exterior of the vehicle body for paint peeling and cracks	Visual			

II. Lubrication, Coolant, Oil, and Water level checks

9	Centralized Lubrication System, Grease Level, if required top up the grease	Visual			
10	Check all Grease Nipple & Fitting	Visual			
11	Power Steering Reservoir Oil level	Visual			
12	Radiator Coolant level	Visual			
13	Expansion tank, Coolant Level	Visual			
14	Wiper Reservoir, Water Level	Visual			
15	Propeller Shaft, Lubricate the grease	Lubrication			

III. Check Driver Cabin,

16	Driver Seat Adjustment/ Seat Belt	Functional			
17	Co-Driver Seat adjustment / Seat Belt	Functional			
18	Inner View Mirror / Clock Time	Functional			

IV. Check After Ignition Key On,

19	Instrument Cluster Warning Symbols & function of all Gauges	Functional			
20	Fault Codes in the Instrument Cluster	Functional			
21	Internal and external lightings	Functional			
22	Check Wiper Motor/ Blade/Arm/Link assembly	Functional			
23	Rear view mirror Position / Function	Functional			
24	Passenger door openings and closing	Functional			
25	Brake Pedal Free Play	Functional			
26	Hand Brake Function	Functional			
27	Suspension Function Check, Kneeling/Normal	Functional			
28	Suspension Function Check, High / Low	Functional			

29	Gear Function Test	Functional			
30	Horn function	Functional			
31	Brake Light and Reverse Light / Warning alarm	Functional			

V. Check Bus Interiors,

32	All Camera Function	Functional			
33	Emergency Alarm/ Panic Switch, Function	Functional			
34	MIC announcement / Speaker	Functional			
35	Charging USB Ports, Passenger seat	Functional			
36	Function of TV and Radio	Functional			
37	Destination Board Mountings & Function	Functional			
38	Passenger Seat Hand Rests	Visual			
39	First Aid box, Fulfilment.	Visual			
40	Roof Emergency Exit (Roof Hatches) both Front & Rear	Visual			
41	Availability of Fire extinguishers and emergency hammers are complete	Visual			

VI. Traction Motor, HV Lines,

42	No coolant leakage at all parts of the Drive Motor and system	Visual			
43	Check the fixing condition of High Voltage line of drive motor	Repair			
44	Check the high voltage harness of the Drive Motor for damage and interference	Repair			
45	Check that all connectors of the Gear shifting mechanism are tightly inserted	Repair			
46	Check & tighten the mounting bolts of Drive Motor / Support Bracket	Repair			
47	Check the Insulation resistance, is within the specified range	Measure			

VII. MCU

48	Check & tightening of mounting bolts of the drive Motor	Repair			
49	Check the connection of high voltage harness and connector of high voltage control cabinet	Repair			
50	Check the wiring harness and connector connection of vehicle controller	Repair			
51	Check the connection of motor controller harness and connector	Repair			
52	Check the Insulation resistance, is within the specified range	Measure			

VIII. Power Battery,

53	Check the connection of the high voltage connecting wire of the power battery	Visual			
54	Check the Insulation resistance, is within the specified range	Measure			
55	Check Coolant leaks from all the Joint	Visual			
56	Surface clean of all HV battery and components	Repair			

IX. Electric Air pump (Compressor),

57	Check Oil is 1 / 2 to 2 / 3 of the sight glass port	Visual			
58	No oil leakage at all parts of the Air pump and the oil seal.	Visual			
59	Check Safety Valve works normally	Visual			
60	Clean the air filter with 4 bar compressed air.	Repair			
61	Check Air Compressor Flexible Pipe (Outlet)	Visual			
62	Check Temperature Sensor Condition	Visual			

X. Electric Steering Pump, Steering System & Tie-Rod,					
63	Check and adjust the free rotation of steering wheel, is $\pm 10^{\circ}$	Functional			
64	Check Steering Wheel Lock nut	Repair			
65	Check all the mounting bolts and Nuts	Functional			
66	No Oil leakage of all parts of electric steering pump.	Visual			
67	Check the Insulation resistance, is within the specified range	Measure			
68	Check Tie rod & Drag rod ball Joints / Dust Cover, Normal clearance $\leq 1.8\text{mm}$	Visual & Functional			
69	Inspect the Tie rod and confirm that there is no Deformation, Bend and Crack	Visual			
70	Inspect all the Fastening bolts and Nuts, confirm that there is no Deformation, Bend and Crack	Visual			
71	Steering rocker Arm, No looseness	Visual			
72	Check and Repair Steering Knuckle Play (King Pin)	Repair			
73	Check Steering Upper and Bottom Drive shaft	Visual			
74	Check King Pin Play, both RHS and LHS	Functional			
XI. Transmission shaft (Propeller Shaft),					
75	Check Fastening bolts & Nuts, No loose	Visual			
76	Check the working condition of universal joint, No Damage/Loose	Visual			
XII. Front and Rear axle,					
77	Inspect "I" Beam and Confirm that there is no Crack	Visual			
78	Front Hub Cup Cover Gasket, Grease Leak (External)	Visual			
79	Rear Hub, Axle Shaft Gasket grease or Oil leak (External)	Visual			
80	Axle Half shaft bolt, not loose	Functional			
81	Check Differential oil leak	Visual			
83	Inspect the Housing and confirm that there is no Deformation, Bend and Crack	Visual			
84	From rear axle, Inspect and confirm that there is no oil leak on the Junction surface between Drive Head and Housing	Visual			
85	Inspect and confirm that, there is no Oil leak on Rear axle Housing Welding line	Visual			
86	Inspect and confirm that, there is no Oil leak on Drain Plug. Form rear axle bottom side.	Visual			
87	Check Diff. Air breather plug should be not lost	Visual			
88	Clean the Diff. Breather Pipe	Repair			
XIII. Brake valve, brake pipeline, ABS system					
89	Inspect the Operating state of Brake Chamber	Visual			
90	Check the working condition of the Brake adjusting mechanism.	Functional			
91	Check Brake Pad thickness	Visual			
92	Brake Clearance Adjustment for all wheels	Functional			
93	Check brake discs and brake callipers	Repair			
94	Check All the pneumatic Valve, Leak	Functional			
95	Check Air leakage in pneumatic system	Visual			
96	Check air pressure regulator Cut out and Cut In pressure	Functional			

97	Check Brake valve and Pipeline connector without Air leakage.	Visual			
98	Check ABS condition, clean ABS ring gear and ABS sensor, etc, by blow air -4 Bar	Clean			
99	Press Brake, check air Leakage / Pressure drop	Functional			

XIV. LV Battery & Fuse Box,

100	Battery Surface is clean	Visual			
101	Check Battery Terminals/ Top bracket, No loose & Distel water level (if required top up)	Functional			
102	Check battery voltage should be over 27V at IC (after start)	Functional			
103	Apply Petroleum Jelly on Terminals	Repair			
104	Check Front & Rear Fuse box and Relays	Visual			

XV. Suspension,

105	Check Air Bellows, Leak	Visual			
106	Check Air Bellows, Heights & adjust if required Front 270 mm /Rear 280 mm	Repair			
107	Remove and check the operation of the shock absorber	Repair			
108	Check Front & Rear Anti Roll / Bushes	Visual			
109	Check All Torsion Bar (Reaction Rod) /Bushes	Visual			

XVI. Tyres,

110	All the Tyre condition incl. Spare wheel (Damage/Under Inflation/Stone remove)	Visual			
111	Check Tyre wear	Visual			
112	Wheel nut need to torque 650NM and Marking required	Repair			

XVII. Others,

113	Electric defroster, Check the Insulation resistance, is within the specified range	Measure			
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XVIII. EV Charging Socket

114	Clean the inner side of the Charging Socket (Contact point) both 1 & 2 & 3.	Clean			
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XIX. Air Condition,

115	Return Grill Air Filter	Clean			
116	Evaporator Air Filter	Clean			
117	Condenser Coil	Clean			
118	Evaporator Coil (90K/180K/270k)	Clean			
119	Check the Control Panel for its normal Operation	Functional			
120	Check abnormal noisy from the A/C system	Functional			
121	Check A/C System for adequate cooling	Functional			
122	Operation of Condenser Fan	Visual			
123	Operation of Evaporator Blower	Visual			
124	Check for any abnormal noisy from Compressor	Visual			
125	Check the refrigerant amount by Sight Glass	Visual			
126	Check the pipe joints for oil stain, it signs of Oil / Gas leakage	Visual			
127	Check all the pipeline is reliably secured	Visual			
128	Check Compressor Mountings	Visual			
129	Inspect all Electric Component mountings	Visual			

XX. Inspect the torque of all the bolts and Nuts

130	Check the torque of the flange and universal joint bolt and lock nut (Propeller Shaft)	Repair			
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131	Check & Torque of slotted nut of left/right knuckle arms.	Repair			
132	Torque of Slotted Nut of Drag link Arms	Repair			
133	Torque of Control Arm Fastening Bolts	Repair			
134	Torque of Slotted Nut of Tie Rod Ball Pins	Repair			
135	Torque of the steering limit bolt lock nut.	Repair			
136	Torque of Lock nut of Front axle Wedged lock Pin (King Pin -Cotter bolt)	Repair			
137	Torque of King Pin cover Bolts, Top and Bottom	Repair			
138	Torque of Brake Disc mounting Bolts	Repair			
139	Check the torque of the disc brake installation bolt / When replace the brake pad	Repair			
140	Torque of bolt and nut of brake chamber & Bracket.	Repair			
141	Check and tight the Front axle I Bolt and Nuts	Repair			
142	Check and Tight, Rear C-Beam Bolts and Nuts	Repair			
143	Check the Torque of Wheel Nuts (No loose)	Repair			

XX. Body and frame,

145	Overall inspection of vehicle body and frame, the body and frame shall be free of fracture, crack, serious rust, deformation, and firm connection.	Visual			
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XXI. Tyre Wear /Pattern/ Depth, Using Depth Gauge.

Sr. No	Tyre Position	Thread Depth mm	Observation /Comments from Tyre Man
1	Front R		
2	Front L		
3	Rear RI		
4	Rear RO		
5	Rear LI		
6	Rear LO		
7	Spare Wheel		

Inspected by - Employ. Name		Supervisor Name	
Signature		Signature	

Sr. No	Comments & Remarks if any